Shirpur Education Society's

R. C. PATEL INSTITUTE OF TECHNOLOGY, SHIRPUR

An Autonomous Institute

(Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere)

आर. सी. पटेल इन्स्टिट्यूट ऑफ टेक्नॉलॉजी, शिरपूर

(स्वायत्त महाविद्यालय)

Programme: B.Tech in Artificial Intelligence (AI) and Data Science

Year: II/Semester III (Exam Year: 2025-2026)

Subject: Data Structures **Max Marks:** 60

Date: 16 Dec 2025 **Time:** 02:30 pm - 04:30 pm (02:00 Hrs.)

END SEMSTER EXAMINATION ODD SEM-III-2025-REGULAR

Instructions:

1. This question paper contains 2 pages
2. Answer to each new question to be started on a fresh page. 3. Figure in right hand side indicates full marks

1. **20**

A. .. **10**

1. Write pseudocode for postfix expression evaluation using stack. Evaluate the following Postfix Expression stepwise using stack operations Expression: 6, 2, 3, + - 3, 8, 2, / + * 2, * 3, + (Note: operands are separated using comma) **10**

---------- OR ----------

2. State key difference between Binary Search Tree and AVL Tree. Create AVL tree from the following operations to organize node in the tree. **10**
   Insert: 67,98,101,70,69, Delete: 101, Insert: 75, 100, 99, 150, Delete: 98
   Traverse organized AVL tree using In-order traversal and write the traversal order of nodes.

B. Apply Merge Sort to sort the array in **descending order.** **10**
Show the recursive division of arrays and the **merge steps** at each level until the array is fully sorted. **Array:** 22, 95, 64, 11, 35, 47, 18, 53, 84, 27, 72, 40 Comment on Complexity of Merge sort

2. **15**

A. a) Apply Fibonacci Search to locate **55** in the following Array C **5**
C = [5, 12, 18, 25, 33, 47, 55, 62]
Show key comparison steps.
b) Write the algorithm/pseudocode for Fibonacci Search.

B. .. **10**

1. A singly linked list contains 6 nodes. You are asked to delete the 4th node. **10**
   The data values are: 5 → 10 → 15 → 20 → 25 → 30 → NULL
   Write the algorithm/pseudocode and show the step-by-step pointer changes after deleting the 4th node.

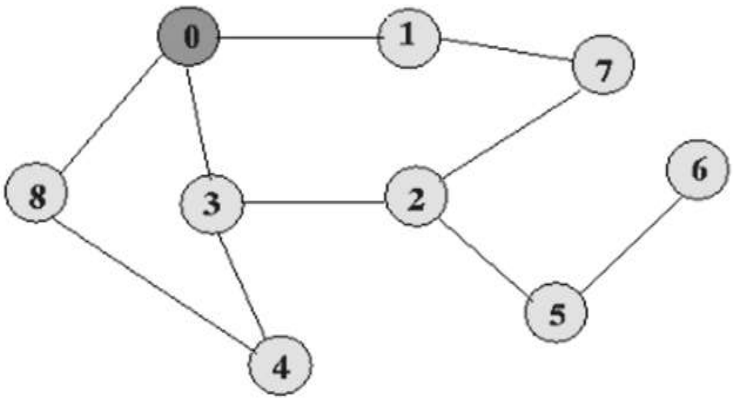
---------- OR ----------

2. a) Apply the concept of BST insertion to construct a Binary Search Tree using the following sequence: 35, 18, 42, 12, 27, 40, 45, 15, 33. Draw the final BST. **10**
   b) Compute the height of the constructed BST.
   c) Write the pseudocode for BST insertion and trace it for inserting 33 into your tree.

3. **10**

A. **5**

Q.1 Understand the given graph, find the depth first search traversal with Starting Node 0. Give the details of each step.



B. .. **5**

1. Explain classification of Data structure and operations of various data structures. **5**

---------- OR ----------

2. Define abstract data type. Explain How List and Stack are Abstract Data Types. **5**

4. **15**

A. .. **10**

1. A doubly linked list stores product ratings in descending order. **10**

Initial List:

99 ⇄ 94 ⇄ 88 ⇄ 85 ⇄ 79 ⇄ 73 head is pointer pointing to first node & value is variable pointing data in node.

Write Pseudocode for the following operations while maintaining descending order:

1. Insert 90 into its correct position
2. Insert 72 at the correct position
3. Delete the node containing 85
4. Delete the node containing 73

Show the resulting list after each operation

---------- OR ----------

2. A smart-city traffic management system needs to handle multiple types of requests in real time. The system receives: **10**

1. Emergency vehicle alerts (Ambulance, Fire Truck, Police) that must be processed with highest priority.
2. Normal vehicle flow updates that can be processed in routine order.
3. Sensor fault alerts that must be added to the front of the queue to be checked immediately, regardless of normal traffic updates.
4. Data upload tasks that arrive continuously and must be stored efficiently without wasting memory

Identify which type of queue (Linear Queue, Circular Queue, Priority Queue, Deque) should be used for each of the above four requirements. For each choice, justify your answer with respect to operations you will perform on each type of Queue

B. What is topological sorting? Discuss topological sort with any suitable example. **5**